# Supplementary Materials

## I. Performance Evaluation and Visualization

Figure 1 illustrates the flow of performance evaluation and visualization evaluation using grid data and sparse data. On the one hand, we directly extract the grid forecasting at sparse locations. Referring to the sparse ground truth, we calculate one-to-many evaluation metrics, which can be fairly compared with the one-to-one evaluation metrics of sparse forecasting. On the other hand, both grid forecasting results and interpolated sparse forecasting results can be visualized. We can choose different subregions for the case study.

## II. Comparison with NWP Models

To demonstrate the importance and irreplaceability of sparse weather forecasting, we compare VN-Net with NWP models at sparse locations. We specifically train models that use historical data from -11h to 0h to forecast specified factors from 1h to 24h for comparison. Figure 2 shows the average metrics of the total of 295 stations among the three regions in China.

Within 24 hours, the overall performance of the three NWP models remains relatively stable. However, NWP models face scale bottlenecks in short-term forecasting and the predictions cannot meet the requirements of accurate sparse forecasting. VN-Net has shown significant advantages in short-term forecasting, particularly in 6-hour and 12-hour. At 18-hour and 24-hour forecasting, VN-Net remains highly competitive in terms of air temperature and wind speed, while it is gradually surpassed by ECMWF in terms of relative humidity.

Compared with NWP models, due to fewer information sources and accumulated errors, the performance of VN-Net gradually declines over time, indicating its insufficient ability in long-term forecasting. However, the low latency and easy accessibility of sparse data perfectly compensate for the mentioned drawback, which means that VN-Net can continuously obtain input data and generate the latest short-term forecasting results.

## III. Case Study

Taking ECMWF as an example, the TIGGE archive provides the 6-hour time resolution forecast of two start times (00:00 and 12:00) per day. In our case study, we choose 6-hour forecasting results for visualization, which means that the time interval between two adjacent forecasting results is 12 hours. Correspondingly, the 6-hour sparse forecasting results are visualized after applying Kriging interpolation.

Several typical cases of different regions and meteorological factors are shown in Figure 3. We present the results of ground truth, ECMWF, N-Net (VN-Net without the visual branch) and VN-Net respectively. By comparing the visualization results, VN-Net exhibits more accurate and consistent forecasting results in all regions and meteorological factors. In particular, VN-Net effectively captures localized meteorological patterns. We mark two red boxes in each set of grid visualization, which are the subregions that differ in visual perception between VN-Net and ECMWF. We further collect a total of 60 forecasting results (a consecutive month) of these subregions and present them in a time-series format. Forecasts generated by VN-Net align more closely with the station observations than those of ECMWF. The case study highlights the ability of VN-Net to deliver accurate and reliable sparse forecasting, which cannot be replaced by grid forecasting models.

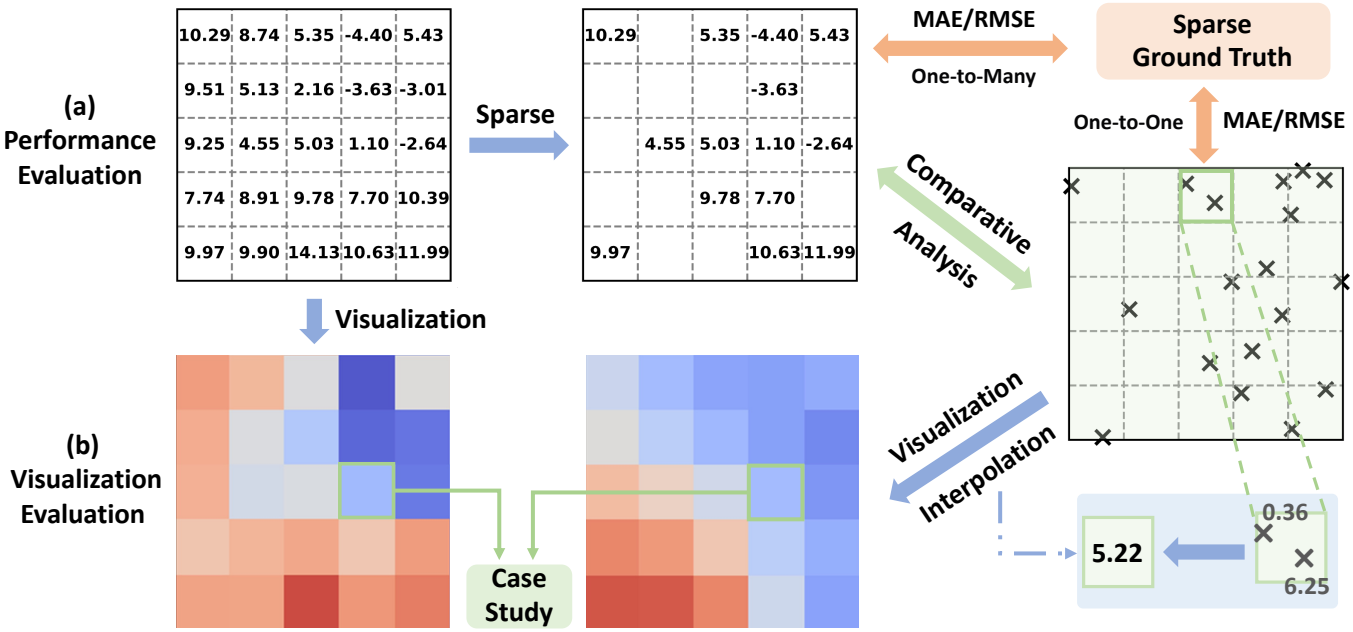


Fig. 1. Illustration of the flow of (a) performance evaluation and (b) visualization evaluation using grid data and sparse data.

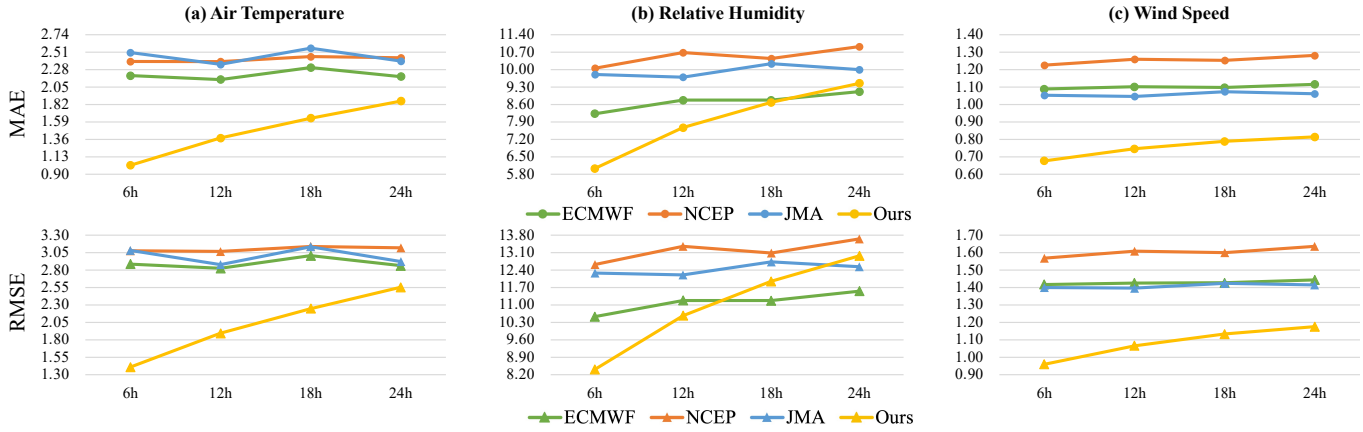


Fig. 2. Comparision with three NWP models in short-term sparse forecasting. Note that our VN-Net only takes 12 hours of historical data as input.

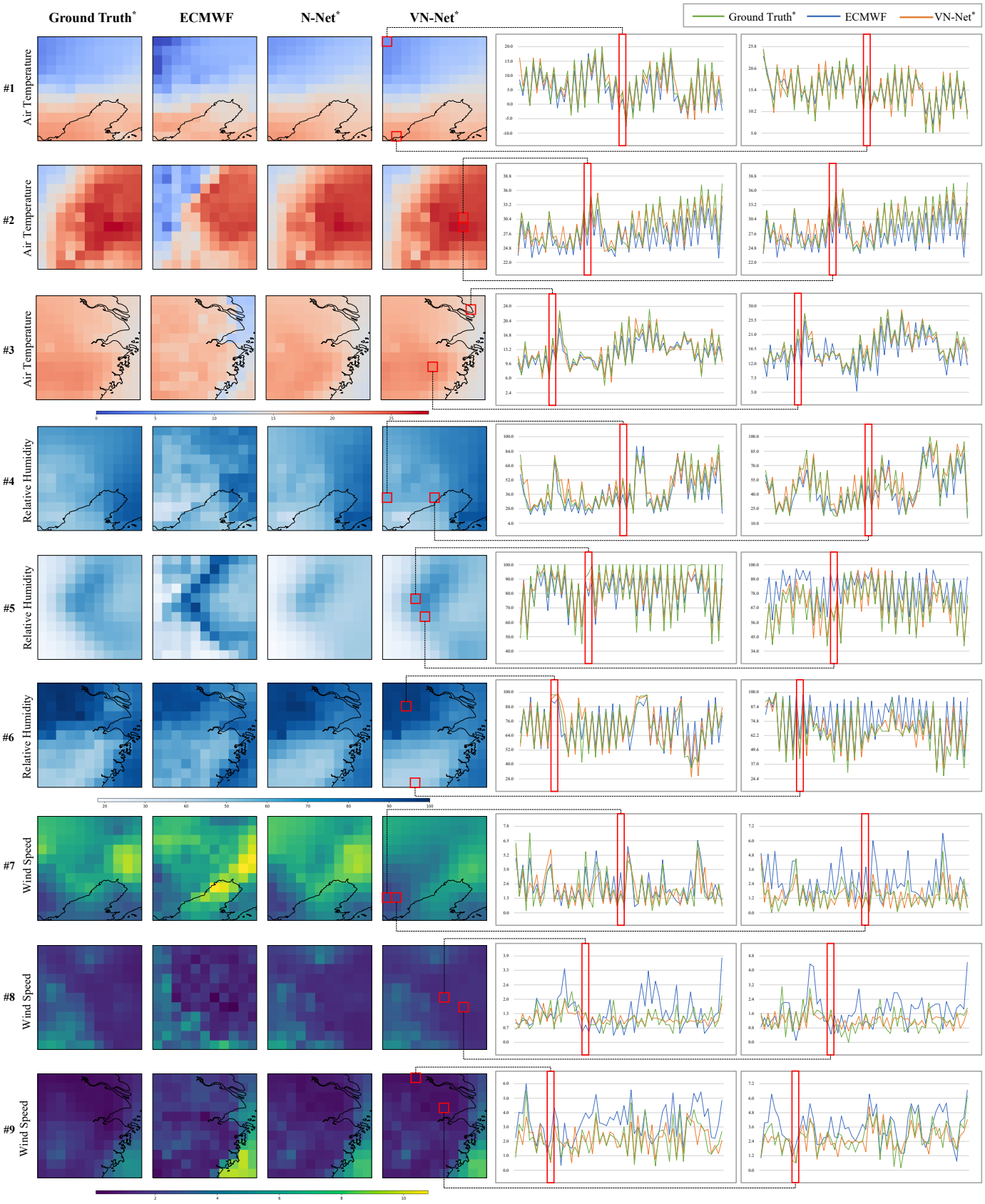


Fig. 3. Case study. Left side: grid visualization of 6-hour forecasting results for different models (#1-#3 air temperature, #4-#6 relative humidity, #7-#9 wind speed). Right side: time-series visualization of selected sparse locations with significant differences (a month, 12 hours per step). N-Net represents VN-Net without the visual branch. * represents the visualization of sparse data to grid data.